A. Madoka and Strange Thoughts

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: math, number theory | https://codeforces.com/problemset/problem/1717/A

Madoka is a very strange girl, and therefore she suddenly wondered how many pairs of integers (a, b) exist, where $1 \le a, b \le n$, for which $(\text{lcm}(a, b))/(\gcd(a, b)) \le 3$.

In this problem, $\gcd(a, b)$ denotes the greatest common divisor of the numbers a and b, and $\text{lcm}(a, b)$ denotes the smallest common multiple of the numbers a and b.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Description of the test cases follows.

The first and the only line of each test case contains the integer n ($1 \le n \le 10^8$).

Output

For each test case output a single integer — the number of pairs of integers satisfying the condition.

Example

Input
6 1 2 3 4 5 100000000

Output
1 4 7 10 11 266666666

Note

For $n = 1$ there is exactly one pair of numbers — (1, 1) and it fits.

For $n = 2$, there are only 4 pairs — (1, 1), (1, 2), (2, 1), (2, 2) and they all fit.

For $n = 3$, all 9 pair are suitable, except (2, 3) and (3, 2), since their lcm is 6, and gcd is 1, which doesn't fit the condition.